Reconstruction gains in antimicrobial peptide generation do not establish generative superiority

Hongsheng Wang[1,2,*] Fei Wu[1] and Feng Lin[3]

[1]College of Computer Science and Technology, Zhejiang University, 866 Yuhangtang Road, Xihu District, 310058, Zhejiang, China
[2]Artificial Intelligence Department, Loctek Ergonomic Technology Corp., Loctek Building, 436 Xueshi Road, Shounan Subdistrict, Yinzhou District, 315100, Zhejiang, China
[3]Fuyao University of Science and Technology, No. 104 Zhihui Avenue, Fuzhou High-tech Industrial Development Zone, 350109, Fujian, China

*Correspondence: Hongsheng Wang. wanghongsheng@zju.edu.cn

Abstract

Reconstruction, sequence-distribution similarity and antibacterial activity are distinct endpoints in antimicrobial peptide generation. We evaluate a reconstruction-objective change within one fixed development protocol. With matched architecture, initialization, batches, decoder and training schedule, a 21-class categorical objective (CE21) raises development residue micro accuracy from 12.55% to 70.90% relative to squared error (MSE21). CE21 improves six specified sequence-distribution distances in both terminal conditions and against both references, but a terminal-conditioned LSTM remains closer on all 24 comparisons. These descriptive results use one training seed and a development set used during model development; fixed KL weighting does not hold effective regularization constant. Seven exploratory routes give diagnostic and gate outcomes that do not establish improved generation. A separate historical laboratory case study includes 30 ordered and 27 assayed peptides. Retrospective MIC aggregation agrees with 105 of 108 reported summary cells. At a strict *S. aureus* threshold below 200 µg/mL, the reported summary gives six active peptides and the retrospective algorithm gives seven; both ordered and assayed denominators are reported. These peptides are not linked to the evaluated models. Comparison with AMPGen illustrates why activity fractions require explicit candidate selection, strain criteria, thresholds, units and denominators. The disclosure framework connects computational endpoints and sample-level laboratory evidence. Reconstruction improvement alone does not establish generative superiority; that claim requires direct generation evaluation and traceable linkage to laboratory evidence.

Keywords antimicrobial peptides, generative models, benchmarking, variational autoencoder, diffusion models, reproducibility

Author description

The authors study machine learning for antimicrobial peptide discovery across computer science, industrial artificial intelligence and biotechnology, emphasizing reproducible evaluation and traceable connections between computational design and laboratory measurements.

Introduction

Many antimicrobial peptides are short cationic sequences studied as templates for antibacterial candidates (50; 51). Variational autoencoders, adversarial networks and diffusion models propose novel peptides (1; 2; 3; 14; 15; 16; 17; 18; 19; 20; 21). Candidate discovery also includes virtual-library enumeration, proteome mining and machine-learning-guided design with laboratory assessment (22; 23; 24). Evaluation practices nevertheless differ in what constitutes evidence of a generative improvement (25).

Three endpoints are routinely reported, and they are not interchangeable. **Reconstruction** measures whether a model can reproduce a known sequence from its latent representation, given the sequence itself as input. **Distributional similarity** measures whether a sampled set resembles a reference set in low-order statistics such as residue and dipeptide composition; metrics of this kind are standard in molecular and sequence generation (26; 27; 28; 29). **Downstream activity** measures whether synthesized material inhibits a target organism. A model can improve the first without improving the second, and can match residue frequencies while producing no useful activity spectrum.

AMP classifiers and activity predictors illustrate the importance of controlled comparison (30; 31; 32; 33; 34; 35; 36; 37). Deep models need not outperform shallow models (4); negative-data selection can bias prediction benchmarks (5), while homology-aware partitioning addresses sequence leakage (38; 39). Related benchmarks (6) and reporting standards (40; 41; 42; 43; 44; 45; 46) motivate explicit control of data exposure and evaluation conditions.

Published AMP pipelines differ in data, terminal-state handling, sampling, selection and assays (1; 2; 3; 7). Recent work covers AI-driven screening (7), generative-model comparisons (8), diffusion design with assays (9), text-guided generation (11), and quantitative MIC and haemolysis benchmarks (13). We ask whether changing

reconstruction loss within one implementation improves reconstruction and sampled-sequence statistics, and whether it exceeds a separately trained comparator.

Training and sampling also differ in variational autoencoders (47), diffusion models (48; 49), protein representation and structure prediction (52; 53), and protein design (54; 55; 56; 57; 58; 59). A training-objective gain requires direct evaluation of the intended sampled-output endpoint.

We compare CE21 and MSE21 under matched architecture, initialization, batches, decoder and training schedule, changing the reconstruction loss applied to the shared output head. We measure both reconstruction and generation endpoints. This isolates the objective change within the fixed protocol, although it does not hold the effective balance between reconstruction and KL regularization constant. A separately trained, terminal-conditioned LSTM provides an executable comparator for generation.

Our contribution is a bounded evaluation of one development protocol. CE21 improves reconstruction and all six specified distributional distances relative to MSE21, but remains behind the LSTM on the 24 metric–terminal–reference comparisons. A separate laboratory record illustrates why assay outcomes alone cannot establish which evaluated model generated an assayed peptide. We use these two records to distinguish measured endpoints from the evidence needed to connect them.

Materials and methods

Study data and analysis scope

The computational study uses the AMP implementation derived from `https://github.com/3DHumanRehab/AMP` (commit `e7486fb`) and the fixed experimental wrappers and outputs in `https://github.com/wanghongsheng01/amp-claude-handoff-20261013` (commit `f1b748d`). Supplementary Methods identify the exact source files and configurations. Analyses use the training and development partitions; the reserved evaluation partition was not accessed.

The accompanying materials contain saved reconstruction and generation outputs, activity predictions, assay readouts and a sample manifest. Independent calculations reproduce the reported computational summaries and compare the original MIC summary with a retrospective algorithm. This recalculation does not repeat model training or laboratory experiments. The historical assay samples have no established checkpoint-to-sample linkage to the evaluated models, and prospective timing of the historical selection and endpoint rules is not established.

Cohort construction and terminal conditions

The cohort derives from a fixed, bounded collection of 512 DBAASP detail cards (10); several other curated AMP repositories exist and differ in admission rules (60; 61; 62; 63). Eligibility required an unchanged canonical 1–48-residue monomer sequence, an unmodified N terminus, a free or amidated C terminus, compatible provider molecular graphs, and traceable bacterial MIC metadata. Sequence grouping used global Needleman–Wunsch identity of at least 70%, counting all alignment columns including gaps, with transitive connected components. Components appearing in the training material were assigned to training or excluded under a fixed exposure rule; all others received a seeded hash assignment with nominal 80/10/10 proportions. Descriptor scaling was fitted on training data only.

The main experiment used a fixed 279-sequence training partition and a 41-sequence development partition, containing 701 non-padding development residues. Free and amidated terminal groups contained 126 and 153 training sequences and 18 and 23 development sequences respectively. Free and amidated sequences are not matched versions of the same sequence, so terminal-group comparisons do not estimate a chemical-modification effect. Terminal state is the only externally supplied condition during generation; no true sequence, residue-aligned descriptor or true length is supplied to any generator. Sequence-derived descriptors remain available to the encoder during reconstruction and training, so reconstruction receives substantially more input information than de novo generation.

The development partition was used during model development and is therefore not an independent confirmation sample. Dataset and code hashes, configurations and execution rules were fixed before the corresponding runs, which does not remove that use.

Reconstruction objectives and decoding

The categorical (CE21) and squared-error (MSE21) models share the same VAE architecture, initialization, batches, random-number schedule, 48-position output, 21-category head and decoding rule. Output categories are the 20 canonical amino acids plus padding. Each VAE has 1,681,549 parameters, including 357 output-head parameters; its diffusion model has 18,612,994 parameters, for 20,294,543 in the two-stage pipeline. A third pipeline replaces the residue-number regression head with a scalar objective and a different normalization, and is reported as a whole-pipeline comparator rather than a single-component ablation.

The two reconstruction objectives differ only in the loss applied to the shared head. Let `s_ijc` be the raw score for category `c` at position `j` in sample `i`, `y_ij` the target including padding, and `B` the batch size:

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{48B}\sum_{i}\sum_{j}\log \operatorname{softmax}(s_{ij})_{y_{ij}}$$

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{96B}\sum_{i}\sum_{j}\left\| s_{ij} - e_{y_{ij}} \right\|^2$$

where `e_y` is the one-hot target. MSE is applied to raw scores, with a factor of one half, and both objectives include padding positions. Both are optimized with the same optimizer, learning rate and update count, so the comparison is component-isolated with respect to the output loss.

Generation from the VAE family follows the fixed sampling path: the diffusion model is trained on the posterior parameters, and a latent is drawn and decoded. A fixed KL coefficient does not control effective regularization strength when the reconstruction loss changes, a limitation we return to in the Discussion.

Autoregressive comparator

The comparator has two LSTM layers (hidden width 64), token embeddings of width 32, terminal embeddings of width 8 and 62,501 parameters. Teacher-forced token training and generation from the terminal condition and sampled prefix use an explicit end-of-sequence token and a 48-residue cap. This locally trained recurrent comparator (1) uses the same data and generation metrics; it does not reproduce a published system's performance.

Training and sampling budgets

Each VAE received 1,000 updates followed by 1,000 matching diffusion updates; the LSTM received 1,000 updates. Each arm used one configuration and one training seed, with no development-selected checkpoint or hyperparameter search within this comparison. Initialization and VAE/LSTM seeds were 222, the diffusion seed 322, and generation seeds 422 and 423 for the free and amidated conditions. VAE and diffusion batch sizes were 32 and 8. Execution was serial on one CPU thread, with common training and generation ceilings of 2,940 s and 1,800 s.

Equal ceilings and update counts do not equalize realized computation, capacity or optimization quality. The smaller LSTM used less computation. Its development negative log-likelihood increased from 2.6153 at update 200 to 3.1698 at update 1000; only the final checkpoint was used.

The CE21 generation step was interrupted after training and completed from the final checkpoint with zero additional training updates. All four methods produced 32 requests each (16 per terminal condition), giving 128 main-comparison outputs; the continuation is not an independent training replicate.

Metrics and interpretation

Reconstruction uses the posterior mean and compares the rendered sequence to each target position; missing predicted positions count as incorrect. Micro accuracy uses all 701 development residues, and amino-acid macro accuracy averages recall over target-supported residue classes. Whole-sequence exact matches and length mean absolute error are also reported, each with 41 sequences as denominator.

We report execution, canonical-alphabet compliance, length eligibility, uniqueness and exact training-copy counts for all requests. Distribution comparison uses canonical sequences of length 1–48, retaining outputs shorter than five residues and those at the cap; a separate length 5–48 eligibility count is reported and does not retroactively filter the main comparison.

Six distances compare each generated terminal group with the corresponding training and development groups: residue and dipeptide Jensen–Shannon divergence in bits, their L_1 distances, length Kolmogorov–Smirnov distance, and biased linear-kernel MMD^2. The latter is the squared distance between mean 421-dimensional features: length/48, 20 residue fractions and 400 dipeptide fractions, with each block weighted by $1/\sqrt{3}$. A one-residue sequence has a zero dipeptide block. JS, L_1 and entropy use pooled counts; MMD averages per-sequence features. Entropy and exact training-copy counts are complementary diagnostics (Supplementary Methods; Tables S1–S2).

All comparisons are descriptive. No generated sequence is treated as an independent training replicate. We report no significance tests or confidence intervals from a single fitted seed, and do not infer antibacterial activity from sequence statistics or treat a nearest training sequence as generation ground truth.

Historical laboratory case study and retrospective MIC analysis

Thirty peptides were ordered and 27 delivered and assayed against *E. coli* ATCC 25922, *P. aeruginosa* ATCC 27853, *S. aureus* ATCC 29213 and *E. faecalis* ATCC 29212. The original 27 × 4 MIC summary is retained unchanged. The assay uses ten two-fold dilutions from 1000 to 1.953125 µg/mL and three replicate wells per concentration, peptide and strain, with ampicillin, polymyxin B and daptomycin as reference antibiotics. These yield 324 replicate dilution series and 3,240 peptide OD_{600} readings. Susceptibility-testing methods and AMP standardization issues (64; 65; 66; 67) motivate explicit reporting of the actual readout and aggregation rule.

For the retrospective algorithm, each replicate endpoint is the lowest tested concentration with $OD_{600} < 0.10$; no qualifying concentration is coded as no endpoint. At least two of three identical numeric calls give a numeric cell endpoint. A majority with no endpoint and a lack of consensus are stored separately; both map to the historical N code only when computing agreement with the original summary. This rule describes retrospective consistency and does not establish the rule used before measurement (Table S5).

Results

Categorical reconstruction substantially improves the reconstruction endpoint

CE21 reconstructs 497 of 701 development residues (70.90%), compared with 88 (12.55%) for MSE21 and 111 (15.83%) for the scalar pipeline (**Table 1**). The CE21–MSE21 difference is 58.35 percentage points under matched capacity, initialization, schedule and decoder. CE21 returns the correct length for all 41 sequences and reproduces three complete sequences.

CE21 macro recall is 42.76%, below its micro accuracy. Eight target-supported classes have zero recall: C, D, E, H, M, N, Q and Y. Improved micro accuracy therefore does not establish balanced residue coverage.

CE21 improves distributional distances but remains behind the recurrent comparator

All eight method–terminal groups produced 16 canonical outputs of length 1–48, unique within each group, with no exact copy of any training sequence. Stricter length 5–48 counts were 14 and 15 for CE21, 16 and 10 for MSE21, and 16 and 16 for both scalar and LSTM, in free and amidated conditions respectively. These describe a finite sample; they do not exclude memorization or imply biological suitability.

CE21 has lower values than MSE21 and scalar on all six sequence-distribution distances, in both terminal conditions and against both references — a consistent component-level improvement. The LSTM, however, has lower values than CE21 on each of those same 24 comparisons (**Figure 1**). Development MMD is 0.017328 for CE21 versus 0.011567 for the LSTM in the free condition, and 0.007710 versus 0.007362 in the amidated condition. The amidated difference is small and receives no significance interpretation.

The ordering is not uniform across all diagnostics. Scalar outputs have smaller absolute training-entropy discrepancies than CE21 for both residue and dipeptide entropy under both terminal conditions, and CE21 has fewer length-eligible outputs than scalar or the LSTM, so neither "all generation metrics improve" nor "all distributions worsen" accurately describes the evidence. The descriptive screen applied to the regenerated samples passed all four of its checks; the generation screen failed eight of 18 checks. Those are decision-rule checks, not eight failed sequences, and they do not alter the status of the interrupted generation step reported in Materials and methods.

Table 1 Development reconstruction from the posterior mean. Counts and errors use 701 target residues or 41 sequences. The autoregressive model has no corresponding reconstruction task.

Model	Correct	Micro (%)	Macro (%)	Length MAE	Exact
CE21	497/701	70.90	42.76	0.000	3/41
MSE21	88/701	12.55	3.57	5.561	0/41
Scalar	111/701	15.83	8.05	1.341	0/41

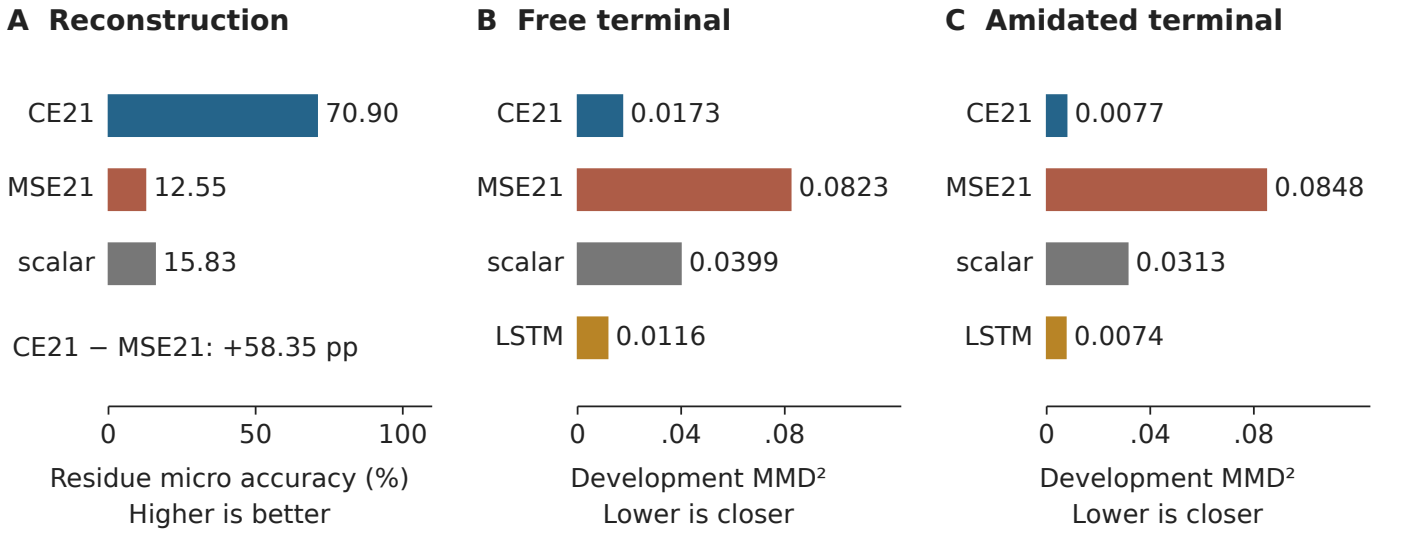


Figure 1 Reconstruction and generation under the fixed development protocol. **(A)** Development residue micro accuracy for the three reconstruction paths. **(B, C)** Development fixed-feature MMD^2 for the four generators in free and amidated terminal conditions; lower values indicate closer feature means. CE21 improves on MSE21, while the LSTM is closest on this generation metric. The scalar pipeline is not a matched single-component control. Results use one fitted seed, 701 reconstruction residues and 16 generation requests per method–terminal group; no uncertainty interval is inferred. Values are recalculated from the saved sequences (Tables S1–S2).

Alt text: Three panels compare reconstruction accuracy and development MMD squared. CE21 reaches 70.90% reconstruction accuracy versus 12.55% for MSE21. LSTM has the lowest generation MMD squared in both terminal conditions.

Seven follow-up routes have distinct diagnostic and gate outcomes

Seven additional diagnostics were run on the same development partition. They investigate the same already-exposed data and are not independent replications.

Training-posterior substitution did not improve development MMD; independent coordinate permutation worsened it. The reference-template prior gave mixed results but used extra template information. These three source diagnostics have no applicable generation winner gate. KL warmup reduced the posterior-mean and eight-draw reconstruction measures from 0.722796 to 0.657852 and 0.661116 to 0.589099, respectively; all four reconstruction/resource guards failed, although its separate investment criterion passed. Latent-conditioned autoregression passed its information gate (eight-draw score 0.690802 with matched latents, 0.097533 with mismatched latents and 0.091958 for a control trained with permuted latent–sequence pairing) but failed the generation gate. Direct-latent diffusion failed its mechanism screen on MMD and dipeptide JS; its value gate was not interpreted. Safe checkpoint loading rejected an unsupported serialized object, leaving that loading check incomplete; this is not evidence of checkpoint corruption. Saved-sequence metric checks passed. Conditioning-context variants were not supported by the multistage screen. Equal-terminal development sequence negative log-likelihoods were 49.16431 (partial), 48.58710 (pooled), 48.91772 (separate) and 48.04302 (fresh autoregressive comparator); 384 attempts produced 381 nonempty and three empty outputs. The historical autoregressive value 57.96692 used a different protocol. Full outcomes and definitions are in Table S4. None of these exploratory results establishes improved generation under the tested protocol (**Figure 2**).

Exploratory activity modelling did not establish a screening model

Two activity-modelling experiments on distinct DBAASP development caches (10) did not establish a screening predictor (Table S3). Native-unit source-group cross-prediction used 46 mass-unit rows in 10 groups and 39 molar-unit rows in seven groups, with 17 folds and 1,700 convolutional updates. Primary MAE for mean, ridge and CNN predictors was 0.811036, 0.775373 and 0.819758 (mass) and 0.516993, 0.582247 and 0.576262 (molar). The convolutional model, motivated by Witten and colleagues (12), did not beat the mean; the mass-unit ridge advantage reversed under alternative weighting. The right-censoring comparison used 173 observations (147 point-valued, 26 right-censored), 34 folds, 102 fits and 588 point-validation predictions. Tail-minus-endpoint primary MAE differences were +0.023094 (mass) and +0.137381 (molar). Between-group mean variance accounted for 90.8% and 71.4% in the mass and molar groups of this censored-cache

Follow-up route	Assessment	Observed outcome
Training-posterior draws	Source diagnostic	MMD not improved
Coordinate permutation	Source diagnostic	MMD worsened
Reference-template prior	Source diagnostic	Mixed; extra information
KL warmup	Reconstruction / resource	4 guards fail
Latent-conditioned AR	Information / generation	PASS / FAIL
Direct-latent diffusion	Mechanism / value	FAIL / not interpreted
Conditioning context	Multistage screen	Not supported

First three routes: no applicable generation winner gate.

KL warmup: its separate investment criterion passes.

Figure 2 Distinct outcomes of seven exploratory routes on the development data. The first three are source diagnostics without an applicable generation winner gate. KL warmup fails four reconstruction/resource guards, while its investment criterion passes. Latent-conditioned autoregression passes the information gate and fails the generation gate. Direct-latent diffusion fails the mechanism gate, leaving its value gate uninterpreted. Conditioning-context variants are not supported by their multistage screen. These are configuration-specific outcomes, not independent replications or model-family verdicts (Table S4).

Alt text: Seven rows list source diagnostics and route-specific outcomes. No common generation gate is assigned to every route. Only latent-conditioned autoregression has an information-pass and generation-fail pair.

diagnostic. Neither activity model selected or validated generated peptides.

Parallel reported and retrospective MIC estimates

The historical case study contains 30 numeric and 78 N cells in the original 108-cell MIC summary; 16 peptides are all-N. The original summary and retrospective algorithm agree in 105/108 cells (97.2%). Table S5 preserves both representations alongside the underlying OD_{600} readouts.

Three endpoints differ: pp1-1/*S. aureus*, 250 versus 125 µg/mL; pp3-14/*P. aeruginosa*, N versus 15.625 µg/mL; and pp1-1/*E. faecalis*, 125 µg/mL versus no endpoint, for reported versus algorithm estimates. These identified differences have not been experimentally adjudicated; the algorithm is not a determination of the true MIC.

Among the 78 reported N cells, 73 have no numeric replicate endpoint, three have one, one has two and one has three. The last is pp3-14/*P. aeruginosa*, whose three replicates yield 15.625 µg/mL. Across all 108 cells, the algorithm gives 30 numeric endpoints, 77 majority-no-endpoint calls and one no-consensus call. N therefore cannot uniformly establish absence of inhibition.

At a strict *S. aureus* MIC threshold below 200 µg/mL, the reported summary gives six active peptides: 6/30 ordered (20%) or 6/27 assayed (22.2%). The algorithm gives seven: 7/30 (23.3%) or 7/27 (25.9%). Neither retrospectively calculated fraction establishes prior commitment to the endpoint.

At strict thresholds below 32, 64, 128 and 200 µg/mL, reported *S. aureus* counts are 6, 6, 6 and 6; algorithm counts are 6, 6, 7 and 7. The original gap from 31.25 to 250 µg/mL does not apply to the algorithm, which includes 125 µg/mL. Activity against any of the four strains below 200 µg/mL gives seven peptides under both representations (**Figure 3**; Table S5).

The reported summary has 11 numeric *S. aureus* and two numeric *P. aeruginosa* entries; the algorithm gives 11 and three, respectively. Figure 3 compares coverage and the *S. aureus* concentration bins without replacing the original summary.

The historical design record describes a 1,000-candidate screen using charge, AMP-prediction score, instability and training-sequence similarity (Supplementary Methods). Thirty peptides were ordered in groups of 8, 8 and 14, labelled broad-spectrum, Gram-positive-specific and Gram-negative-specific; the corresponding assayed counts were 7, 8 and 12. The unassayed identifiers are pp1-6, pp3-8 and pp3-10; their absence does not establish a synthesis-failure cause.

Under the historical operational spectrum-label rule, agreement is 2/8, 1/8 and 0/14 by ordered group, or 3/30 ordered and 3/27 assayed overall, under both MIC representations. This differs from *S. aureus* activity and does not establish biological specificity beyond the four-strain panel; N is treated only as failure to meet the numeric condition.

Purity certificates show 26 peptides at 95.0–99.1% and pp1-7 at 80.3%, meeting its ≥80% order specification. Ten peptides are recorded as insoluble in water and 18 in DPBS; these certificates do not

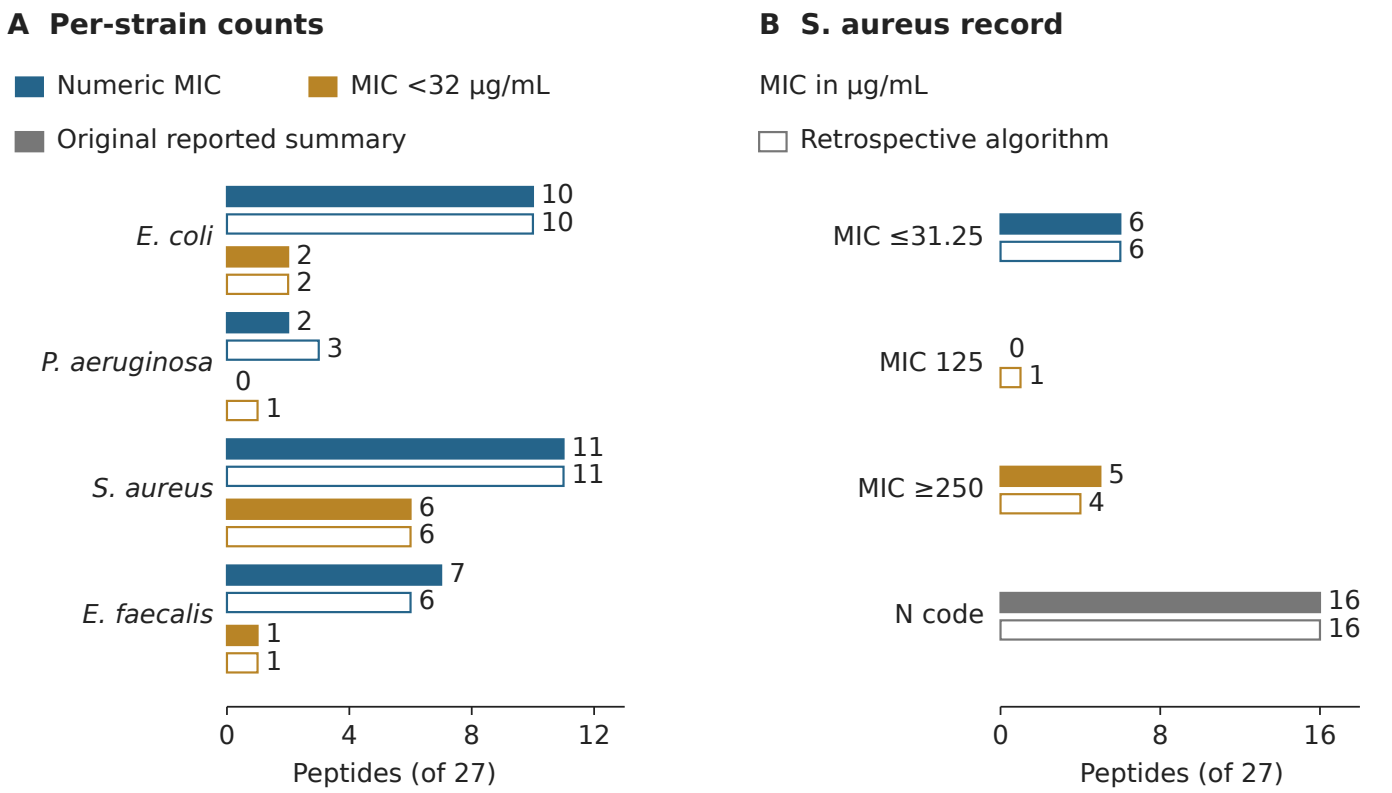


Figure 3 Reported and retrospective-algorithm MIC estimates for 27 assayed peptides. **(A)** Numeric MIC counts and counts strictly below 32 µg/mL for each strain, with both representations shown. **(B)** *S. aureus* distributions: the reported summary contains six values at or below 31.25, five at or above 250 µg/mL and 16 N values; the algorithm contains six, four and 16, respectively, plus one value at 125 µg/mL. N combines majority-no-endpoint and no-consensus states only for historical comparison. The algorithm does not adjudicate true MIC values.

Alt text: Filled and outlined bars compare reported and algorithm endpoints by strain. The *S. aureus* distribution changes from five values at or above 250 to four, with one value at 125; its count below 200 changes from six to seven.

establish the assay solvent. Structural predictions are not experimental structures (68; 69). Haemolysis and cytotoxicity were not assayed; haemolysis predictors (70; 71; 72) do not supply such measurements.

Comparison with AMPGen under different strain and endpoint definitions

AMPGen designed 40 peptides and synthesized 38; 31/38 (81.58%) had MIC ≤75 µM against *E. coli* K88 or *S. aureus* ATCC 29213 (9). Nine peptides with MIC < 5 µM against both initial targets were subsequently tested against *E. coli* ATCC 25922, *P. aeruginosa* ATCC 27853, *S. aureus* ATCC 25923 and *E. faecalis* ATCC 29212. Thus, the 31/38 fraction does not come from the four-strain extension, and its *S. aureus* extension strain differs from this case study.

The studies use different candidate-selection procedures, strain criteria, thresholds and concentration units. Converting 75 µM using the 27 present-case molecular weights gives 80.27625–243.51975 µg/mL, illustrating peptide-dependent conversion rather than AMPGen measurement values. Table S6 distinguishes AMPGen's fraction from this case study's reported and algorithm activity fractions and operational spectrum-label agreement.

The differing numerator and denominator definitions are distinct estimands. A harmonized comparison requires peptide-level measurements and common inclusion and activity rules; these data do not assign differences in reported fractions to model performance, biology or criterion choice.

Discussion

Three endpoints, three different claims

The central empirical finding is that a reconstruction gain of 58.35 percentage points accompanies improved distributional distances over MSE21, but does not exceed the executable LSTM comparator on those distances. The different information available during reconstruction and generation provides a possible explanation, but the present comparisons do not identify the causal mechanism. Reconstruction receives the input sequence and residue-aligned descriptors; sampling receives a terminal condition only.

The latent-conditioned diagnostic shows decodable cached sequence information (0.690802 matched versus 0.097533 mismatched), yet its generation gate fails. It does not isolate optimization, latent-distribution mismatch, decoder behaviour, loss scaling or limited

data as the cause. Fixed KL weighting also does not fix effective regularization when reconstruction-loss scaling changes.

Interpreting the negative follow-up results

The seven routes examine latent substitution, prior information, permutation, regularization, decoder conditioning and diffusion targets. Their distinct diagnostics and gates are reported in full, including the incomplete safe-loading check for direct-latent diffusion. These configuration-specific outcomes on exposed development data do not exhaust possible improvements.

What the wet-laboratory record establishes

The historical case study supports endpoint comparison from raw measurements, not attribution of laboratory activity to the evaluated generators. Agreement of 105/108 cells and the parallel six-versus-seven activity counts quantify retrospective consistency. Neither establishes prospective endpoint selection or experimentally adjudicates the three differences.

Raw readings enable retrospective analysis. Linking a physical sample to a generator additionally requires peptide identifiers tied to a checkpoint, configuration and sampling run; assay activity alone cannot establish that link.

We propose six disclosures for an AMP generative claim invoking laboratory evidence (**Box 1**). **Figure 4** maps them to this case study without assigning an aggregate score.

Box 1. Minimum disclosure for a laboratory-validated AMP generation claim.

1. **Selection protocol, fixed in advance** — pool size, scoring function, selection rule, and the exact stage at which each peptide was chosen.
2. **Checkpoint-to-sample linkage** — model identifier, checkpoint hash, configuration hash and seed for every assayed peptide.
3. **Denominator and threshold, fixed in advance** — the strain set, the activity threshold with units, and whether the rule is strict or inclusive, stated before measurement.
4. **Replicate semantics** — technical or biological replication, the aggregation rule, and distinct encodings for no endpoint and no consensus.
5. **Per-sample raw readouts** — measured values with units, plus explicit encoding of non-numeric outcomes.
6. **Independence statement** — whether the evaluation data were used during model development, and whether the comparator models were tuned with an equivalent budget.

The historical selection rules and the retrospective denominator and threshold definitions are stated, but prospective timing is not established. Sample identifiers, raw OD values and both MIC representations are provided; the three differing endpoints remain unadjudicated. Sample-to-checkpoint attribution is not established. Development-data exposure and comparator budgets are disclosed separately and do not establish independent evaluation or equivalent optimization.

Relation to prior evaluation work in this journal

The evaluation follows endpoint-focused AMP-classification critiques (4; 5) and the GPCR–peptide binding benchmark (6), applied here to generation. The matched objective change improves reconstruction and distributional distances over its control but remains behind a separate comparator.

The AMPGen comparison illustrates why overlapping strain identities across different experimental stages do not harmonize success fractions. A generative-superiority claim requires direct generation evaluation, an executable comparator and explicit tuning and data-exposure conditions.

Limitations

The comparison uses one training seed, 16 requests per terminal group and a repeatedly accessed development set, not independent validation. Final-checkpoint LSTM optimization is not established as adequate; further tuning could change the ranking in either direction. Capacity and realized computation differ, and the VAE pipelines additionally train diffusion models. Completing interrupted CE21 generation does not add a replicate.

Activity-modelling caches are small, heterogeneous and exposed. The historical assays have no established attribution to an evaluated model, three unadjudicated endpoint differences, and unverified prospective timing of selection and endpoint rules. The study does not establish biological safety, a general ranking of model families or unsuitability of a model class for peptide design.

Conclusion

In the fixed 279-training/41-development-sequence protocol, CE21 improves reconstruction from 12.55% to 70.90% and all six specified distances over MSE21, while LSTM remains closer on all 24 generation comparisons. These findings concern one seed and exposed development data. The seven exploratory routes do not establish improved generation under their respective diagnostics and gates.

The separate historical case study gives six reported and seven algorithm-derived *S. aureus*-active peptides at the stated threshold. The disclosure framework distinguishes reconstruction, sampled-distribution similarity and laboratory activity, and the sample-level provenance needed to connect those endpoints.

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Author contributions

Hongsheng Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Visualization.

Fei Wu: Conceptualization (academic supervision), Writing – Review & Editing, Supervision, Project administration, Funding acquisition.

Feng Lin: Resources; Conceptualization (industrial scenario and requirements); Writing – Review & Editing; Supervision (industrial supervisor).

Fei Wu is the guarantor of the work.

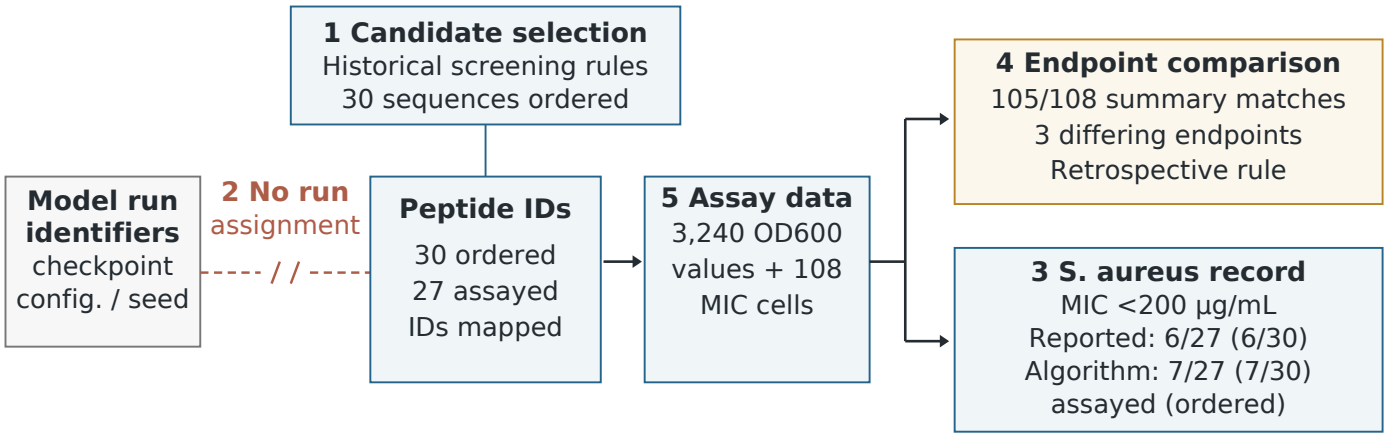


Figure 4 Evidence chain for interpreting the historical laboratory case study. Numbers correspond to Box 1. The historical selection description (1) does not establish decision timing. Sample identifiers and 3,240 raw OD_{600} values (5) support retrospective endpoint comparison (4): 105/108 MIC cells agree and three differ. The reported and algorithm *S. aureus* counts below 200 µg/mL are six and seven (3), each using ordered and assayed denominators. These calculations do not adjudicate laboratory MICs. No checkpoint-to-sample assignment is established (2). The computational comparison's data exposure and budgets are disclosed separately (6).

Alt text: A laboratory evidence chain connects 30 ordered samples, 27 assayed samples and 3,240 optical-density readings to parallel MIC summaries. A missing assignment link separates these samples from model checkpoints. A separate computational panel identifies 279 training and 41 development sequences, one seed and four generators.

Competing interests

Hongsheng Wang's affiliation with Loctek Ergonomic Technology Corp. and the institutional collaborations are disclosed in the author list. The authors report no competing interests beyond these disclosed collaborations.

Relationship to prior work

Hongsheng Wang and Feng Lin are also authors of AMPGen (9). AMPGen reports a generator with laboratory evaluation; the present study evaluates a reconstruction-objective change and uses a separate historical assay case study. Shared reference-antibiotic and strain identities do not establish comparable study endpoints.

Ethics statement

No human participants, human tissue or animal subjects were involved in this work. All bacterial work used the standard reference strains described in Materials and methods.

Data and code availability

Saved reconstruction and generation sequences, activity predictions, frozen configurations, analysis code, raw OD_{600} readings, original and retrospective MIC estimates, and sample and purity manifests accompany this manuscript as reviewer materials. Supplementary Tables S1–S6 report the corresponding summaries. Public versioned materials: `https://github.com/wanghongsheng01/bib-amp-review-materials-20261023/tree/v3.1-2026-10-23`; materials DOI: `https://doi.org/10.5281/zenodo.23586233`. The accompanying offline scripts recalculate summary results without model training or access to the reserved evaluation partition. Source repositories and versions are identified in Materials and methods.

Acknowledgements and use of generative AI tools

The authors wrote the manuscript. OpenAI Codex assisted with source and reference checking, code inspection, independent recalculation of saved outputs, deterministic plotting code, factual corrections and language editing of the author-written text. No new laboratory measurements or generative images were created. Human authors retain responsibility for the scientific content. Supplementary Methods record the assistance used in this revision.

Supplementary material

Supplementary Methods and Tables S1–S6 contain reconstruction results, generation distances, activity-regression outputs, route evidence, sample-level MIC comparisons and the AMPGen endpoint comparison. The accompanying offline reviewer materials provide the underlying data and calculation scripts.

Key Points

- CE21 raises development residue micro accuracy from 12.55% to 70.90% and improves six specified generation distances over the matched MSE21 control.
- A terminal-conditioned LSTM remains closer on all 24 descriptive comparisons; the study uses one training seed and exposed development data.
- Seven exploratory routes have distinct diagnostic and gate outcomes and do not establish improved generation under the tested configurations.
- In a separate 30-ordered/27-assayed laboratory case study, reported and retrospective MIC estimates agree in 105/108 cells; the stated *S. aureus* activity count is six or seven, respectively.
- Sample-to-checkpoint attribution, endpoint definitions and data exposure must be stated before laboratory activity can support a claim about an evaluated generator.

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